Chapter 2. Supervised Learning and Model Assessment

Lecture 4 (regression function, parametric vs nonparametric, KNN, bias–variance, adjusted R^2, AIC, BIC)

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1 The Supervised Learning Setup

Motivation. $Y = f(X) + \epsilon$ separates what can be learned (f) from what cannot (ϵ), so even a perfect method has nonzero test error.

Example: advertising data

`Sales` and advertising budgets `TV`, `Radio`, `Digital` in many markets: $Y =$ `Sales`, $X = (X_1, X_2, X_3)$,

$$Y = f(X) + \varepsilon,$$

f unknown, $\mathbb{E}\varepsilon = 0$, ε independent of X. Similarly: income of 30 individuals versus education, seniority and marital status.

1.1 Why estimate f?

Prediction: $\hat{f}(x)$ at a new $X = x$. **Inference**: which predictors matter, and the direction and size of their effects (e.g. the change in sales from an extra \$1,000 on TV).

1.2 The regression function

At $X = 4$ many Y values are possible; the best prediction is:

Theorem: the conditional mean is the optimal predictor

$$f(x) = \arg\min_g \mathbb{E}[(Y - g(X))^2 \mid X = x] = \mathbb{E}(Y \mid X = x),$$

called the **regression function**. Moreover, for any g, $\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - f(X))^2] + \mathbb{E}[(f(X) - g(X))^2]$.

Proof [Supplement]. For a constant c, $\mathbb{E}[(Y - c)^2 \mid X = x] = \mathrm{Var}(Y \mid X = x) + (f(x) - c)^2$, minimized at $c = f(x)$. For the decomposition, write $Y - g = (Y - f) + (f - g)$; the cross term vanishes because $\mathbb{E}[(Y - f(X)) \mid X] = 0$. ■

$\mathrm{Var}(Y \mid X = x) = \mathrm{Var}(\varepsilon)$ is the **irreducible error**; $\mathbb{E}(f - g)^2$ is the **reducible error**.

2 Parametric Approach

1. specify a functional form, e.g. the linear model $f_L(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$;
2. estimate the parameters $(\beta_0, \ldots, \beta_p)$ from data (e.g. by least squares).

A low-dimensional parameter space gains interpretability and stability but risks **model misspecification**. More parameters, more flexibility: $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ versus $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$.

3 Nonparametric Approach

3.1 Local averaging

Few observations have exactly $X = 4$, so we average nearby points:

$$\hat{f}(x) = \text{Ave}(Y \mid X \in \mathcal{N}(x)),$$

with $\mathcal{N}(x)$ a neighborhood of x.

Definition: K-nearest neighbors regression

$$\hat{f}(x) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} y_i,$$

where $\mathcal{N}_K(x)$ is the set of the K training points closest to x.

Others: kernel regression, smoothing splines (Ch. 7), Gaussian processes. $K = 1$ interpolates the data; $K = 9$ is smoother.

Theorem: bias and variance of KNN at a point

Let $Y = f(X) + \varepsilon$ with $\text{Var}\varepsilon = \sigma^2$ and condition on the inputs. With neighbors $x_{(1)}, \ldots, x_{(K)}$ of x_0,

$$\mathbb{E}\big[(Y_0 - \hat{f}(x_0))^2\big] = \sigma^2 + \Big[f(x_0) - \frac{1}{K}\sum_{l=1}^{K} f(x_{(l)})\Big]^2 + \frac{\sigma^2}{K}.$$

Proof [Supplement]. $\hat{f}(x_0) = \frac{1}{K}\sum_l f(x_{(l)}) + \frac{1}{K}\sum_l \varepsilon_{(l)}$; its variance is σ^2/K and its mean is the average of f over the neighbors. Apply the bias–variance decomposition (§4.3). ■

Small K: low bias, high variance σ^2/K; large K: higher bias. *Smaller K = more* flexible.

3.2 Pros and cons

No assumption on the form of f, but a very large n is needed (curse of dimensionality, Ch. 1), and a very flexible fit can reproduce the training data and still predict badly.

4 Assessing Model Accuracy

4.1 Complexity, underfitting, overfitting

Underfitting: too simple a model, systematic errors. **Overfitting**: the model fits noise rather than signal.

4.2 Training error and test error

For a model fitted on $\text{Tr} = \{(x_i, y_i)\}_{i=1}^n$,

$$\text{MSE}_{\text{Tr}} = \text{Ave}_{i \in \text{Tr}}\big[y_i - \hat{f}(x_i)\big]^2, \qquad \text{MSE}_{\text{Te}} = \text{Ave}_{i \in \text{Te}}\big[y_i - \hat{f}(x_i)\big]^2,$$

with $\text{Te} = \{(x_i, y_i)\}_{i=1}^M$ fresh observations. Training error is optimistic, especially for flexible models. Simulations:

(1) moderately curved truth: training MSE falls monotonically, test MSE is U-shaped;

(2) nearly linear truth: the linear model and a smooth fit do best;

(3) wiggly truth, low noise: very flexible fits do best.

4.3 Bias–variance trade-off

Theorem: decomposition of the expected test error

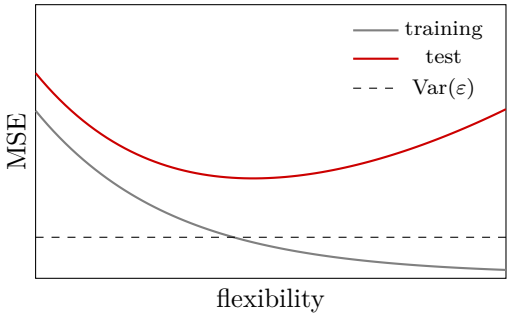
Let (x_0, y_0) be a new observation, $y_0 = f(x_0) + \varepsilon_0$, $f(x) = \mathbb{E}(Y \mid X = x)$, and let $\hat{f}$ be fitted on an independent training set. Then

$$\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{[\mathrm{Bias}(\hat{f}(x_0))]^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{irreducible}} .$$

Proof [Supplement]. $y_0 - \hat{f} = \varepsilon_0 + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$; the three terms are uncorrelated (ε_0 is independent of the training data and has mean 0; the second term is constant; the third has mean 0). Square and take expectations. ■

More flexibility: lower bias, higher variance.

(a) training vs. test error



(b) decomposition of test MSE

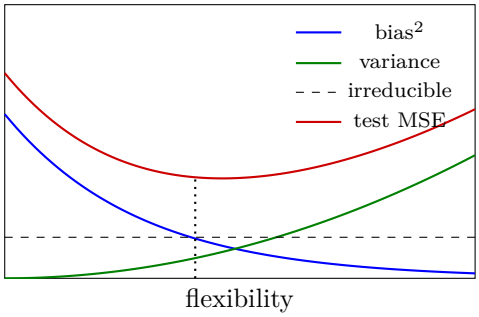
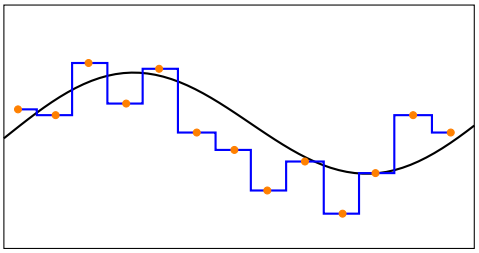


Figure 1: Bias–variance trade-off (schematic); dotted: optimal flexibility.

$K = 1$: low bias, high variance



$K = 9$: smoother, more bias

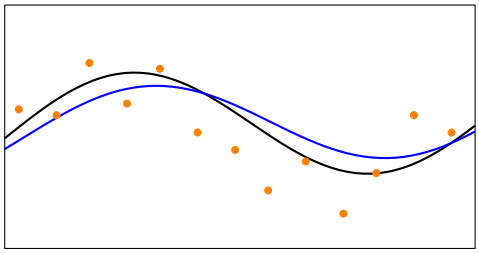


Figure 2: KNN regression (schematic). Black: true f; blue: fit.

5 Criteria Based on the Training Data

$$\text{Score} = \text{goodness of fit} + \text{complexity penalty}, \qquad \mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

5.1 Adjusted R^2

$R^2 = 1 - \text{RSS}/\text{TSS}$ always increases when a predictor is added;

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}.$$

It rises iff $\text{RSS}/(n-p-1)$ falls, i.e. iff the new predictor has $|t| > 1$ (Exercise 2.2).

5.2 AIC and BIC

For a Gaussian linear model with p predictors,

$$\text{AIC} = n\log(\text{RSS}/n) + 2(p+1), \qquad \text{BIC} = n\log(\text{RSS}/n) + \log(n)(p+1);$$

smallest value wins. In general, with k parameters, $\text{AIC} = -2\log L(\hat\theta) + 2k$, $\text{BIC} = -2\log L(\hat\theta) + k\log n$.

Derivation: information-theoretic interpretation of AIC

$\text{KL}(p_0 \| p_\theta) = \mathbb{E}_{p_0} \log \frac{p_0(X)}{p_\theta(X)}$ equals, up to a constant, $-\mathbb{E}_{p_0} \log p_\theta(X)$, the expected log-loss on a *new* observation; $\log L(\hat\theta)$ is optimistic since $\hat\theta$ was fitted to the same data. A second-order Taylor expansion around θ^* (correct model, regular case): fitting and evaluating on new data each contribute about $k/2$, so

$$\mathbb{E}\big[-2\,\mathbb{E}_{\text{new}} \log p_{\hat\theta}(X^{\text{new}})\big] \approx \mathbb{E}\big[-2\log L(\hat\theta)\big] + 2k.$$

so AIC estimates the expected out-of-sample deviance (Akaike).

Derivation: Bayesian interpretation of BIC

With equal prior weights, $P(M_k \mid \text{data}) \propto \int L_k(\theta)\pi_k(\theta)d\theta$; a Laplace approximation around $\hat\theta_k$ gives

$$\log P(M_k \mid \text{data}) \approx \log L(\hat\theta_k) - \frac{k}{2}\log n + O(1)$$

(Schwarz, 1978); times -2 this is BIC, so the smallest BIC has the largest posterior probability.

	Penalty	Goal	Model size	Consistent?
AIC	$2k$	prediction	larger models	no
BIC	$k\log n$	true-model selection	smaller models	yes

BIC penalizes more once $n \geq 8$ ($\log n > 2$). **Consistency**: if the true model is a candidate, BIC selects it with probability $\to 1$; AIC keeps a positive probability of overfitting.

Summary

- $\mathbb{E}(Y \mid X = x)$ is optimal under squared loss; $\text{Var}(\varepsilon)$ is irreducible.
- Parametric: interpretable, may be misspecified; KNN-type local averaging: flexible, data-hungry.
- Test error is U-shaped in flexibility (bias–variance); AIC targets prediction, BIC consistency.

6 Selected exercises

Motivation. Three derivations the chapter uses without proof.

Exercise 2.1 (Absolute loss and the conditional median)

(a) Let Y be a random variable with $\mathbb{E}|Y| < \infty$ and let m be a median, $\Pr(Y \le m) \ge \frac{1}{2}$ and $\Pr(Y \ge m) \ge \frac{1}{2}$. Show that $\mathbb{E}|Y - c| \ge \mathbb{E}|Y - m|$ for every $c \in \mathbb{R}$.
(b) Deduce that $\mathbb{E}|Y - g(X)|$ is minimized over functions g by a conditional median $g(x) = \text{median}(Y \mid X = x)$.

Solution [Supplement].

(a) All expectations below are finite because $\mathbb{E}|Y| < \infty$.

Step 1 ($c > m$). If $y \le m$, then $|y - c| - |y - m| = (c - y) - (m - y) = c - m$. For every y, the triangle inequality gives $|y - c| - |y - m| \ge -(c - m)$. Hence

$$\mathbb{E}\big[|Y - c| - |Y - m|\big] \ge (c - m)\Pr(Y \le m) - (c - m)\Pr(Y > m) = (c - m)\{2\Pr(Y \le m) - 1\} \ge 0.$$

Step 2 ($c < m$). Symmetrically, $|y - c| - |y - m| = m - c$ for $y \ge m$ and $\ge -(m - c)$ always, so the difference has expectation $\ge (m - c)\{2\Pr(Y \ge m) - 1\} \ge 0$.

(b) Let $m(x) = \inf\{c : \Pr(Y \le c \mid X = x) \ge \frac{1}{2}\}$ (a measurable conditional median). By (a) applied conditionally, $\mathbb{E}\big[|Y - g(X)| \,\big|\, X = x\big] \ge \mathbb{E}\big[|Y - m(X)| \,\big|\, X = x\big]$ for every x; take expectations over X. ■

Exercise 2.2 (Adjusted R^2 rises iff $|t| > 1$)

Let M_0 have design $X_0 \in \mathbb{R}^{n \times (p+1)}$ (intercept and p predictors) and residual sum of squares RSS_0, and let M_1 add one predictor z, with design $X_1 = [X_0, z]$ of full column rank $p + 2 < n$ and $\text{RSS}_1 > 0$. Let $t = \hat\gamma / \text{SE}(\hat\gamma)$ be the usual t-statistic of the coefficient $\hat\gamma$ of z in M_1, with $\hat\sigma^2 = \text{RSS}_1/(n - p - 2)$.
(a) Show that $\text{RSS}_0 - \text{RSS}_1 = t^2\,\text{RSS}_1/(n - p - 2)$.
(b) Show that $R^2_{\text{adj}}(M_1) > R^2_{\text{adj}}(M_0)$ if and only if $|t| > 1$.

Solution [Supplement].

(a) Let H_0 be the orthogonal projection onto $\text{col}(X_0)$ and $\tilde z = (I - H_0)z$; $\tilde z \neq 0$ since X_1 has full column rank.

Step 1. $\text{col}(X_1) = \text{col}(X_0) \oplus \text{span}(\tilde z)$ with $\tilde z \perp \text{col}(X_0)$, so the projection onto $\text{col}(X_1)$ is $H_1 = H_0 + \tilde z \tilde z^\top / (\tilde z^\top \tilde z)$.

Step 2. Since $\tilde z^\top H_0 = 0$, we have $\tilde z^\top y = \tilde z^\top (I - H_0) y$ and

$$\text{RSS}_1 = y^\top (I - H_1) y = \text{RSS}_0 - \frac{(\tilde z^\top y)^2}{\tilde z^\top \tilde z}.$$

Step 3. Put $c = \tilde z^\top y / \tilde z^\top \tilde z$. Then $H_1 y = H_0 y + c\tilde z = (H_0 y - cH_0 z) + cz$, where $H_0 y - cH_0 z \in \text{col}(X_0)$. The coefficients of $H_1 y$ in the basis given by the columns of X_1 are unique, so $\hat\gamma = c$. As a linear function of y, $\text{Var}(\hat\gamma) = \sigma^2 / \tilde z^\top \tilde z$, so $\text{SE}(\hat\gamma)^2 = \hat\sigma^2 / \tilde z^\top \tilde z$.

Step 4. Hence $t^2 = \dfrac{(\tilde z^\top y)^2}{\tilde z^\top \tilde z\, \hat\sigma^2} = \dfrac{\text{RSS}_0 - \text{RSS}_1}{\hat\sigma^2}$, which is (a).

(b) Both models share TSS, so M_1 has the larger R^2_{adj} iff $\text{RSS}_1/d < \text{RSS}_0/(d+1)$ with $d = n - p - 2$. By (a), $\text{RSS}_0 = \text{RSS}_1(1 + t^2/d)$, so the condition reads $d + 1 < d + t^2$, i.e. $t^2 > 1$. ■

Exercise 2.3 (The Gaussian log-likelihood in AIC and BIC)

Let $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n\times(p+1)}$ of full column rank, and assume $\mathrm{RSS} > 0$ for the least-squares fit.
(a) Show that $\max_{\beta,\sigma^2} \log L(\beta, \sigma^2) = -\frac{n}{2}\{\log(2\pi) + \log(\mathrm{RSS}/n) + 1\}$.
(b) Deduce that counting σ^2 as a parameter ($k = p + 2$) changes AIC and BIC only by a constant common to all models fitted to the same n observations, so the model rankings agree with the formulas of this chapter.

Solution [Supplement].

(a)

Step 1. $\log L = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - X\beta\|^2$. For each fixed σ^2 this is maximized by minimizing $\|y - X\beta\|^2$, i.e. at the least-squares $\hat\beta$, giving $h(s) = -\frac{n}{2}\log(2\pi s) - \mathrm{RSS}/(2s)$ with $s = \sigma^2$.

Step 2. $h'(s) = -\frac{n}{2s} + \frac{\mathrm{RSS}}{2s^2} = \frac{\mathrm{RSS} - ns}{2s^2}$ is positive for $s < \mathrm{RSS}/n$ and negative beyond, so h is maximized at $\hat\sigma^2 = \mathrm{RSS}/n$, where $h = -\frac{n}{2}\log(2\pi\,\mathrm{RSS}/n) - \frac{n}{2}$.

(b) By (a), $-2\log L(\hat\theta) = n\log(\mathrm{RSS}/n) + n(1 + \log 2\pi)$. With $k = p + 2$,

$$\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2(p+1) + \{n(1+\log 2\pi) + 2\},$$
$$\mathrm{BIC} = n\log(\mathrm{RSS}/n) + (p+1)\log n + \{n(1+\log 2\pi) + \log n\}.$$

The braces do not depend on the model, so differences between models, and hence the selected model, are unchanged. ■